First brain *de novo* transcriptome of the Tyrrhenian tree frog, *Hyla sarda*, for the study of dispersal behavior

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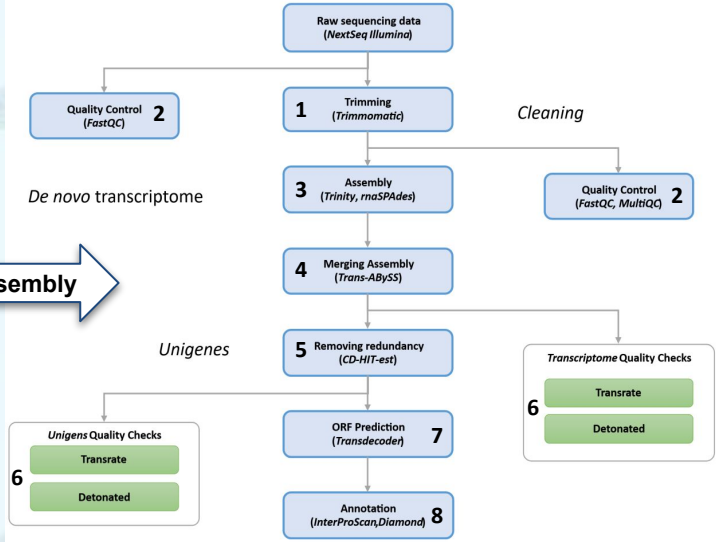
Background
Nine individuals of *Hyla Sarda* species were selected representing two distinct behavioural profiles: bold and shy exploratory behaviour in a new environment. The brains were dissected and immediately preserved in RNAprotect Tissue Reagent until RNA extraction

Sample Code	Phenotype	Raw sequences	% Trimmed reads
HS9	Bold	72098178	96.4
HS 17	Bold	53786371	96.7
HS 27	Bold	54282464	96.3
HS 44	Bold	49300066	96.6
HS 56	Bold	51227766	91.0
HS 87	Shy	53775786	94.0
HS 93	Shy	54008324	93.5
HS 104	Shy	51829828	91.9
HS 111	Shy	52754498	93.2

https://www.ncbi.nlm.nih.gov/sra/?term=PRJEB51203

Assembly



Methods
De novo transcriptome assembly involves:
1.Trimming phase
2. Quality control step
3. Use of two different assemblers
4. Merging phase (unique assembly consensus file)
5. Removing transcript redundancies
6. Transcriptome validation.
7. ORF prediction
8. Transcriptome annotation (both by homology and functional).

Results

Annotation results:
The contigs were aligned with DIAMOND on NCBI nr, Swiss-Prot and TrEMBL to retrieve the best matching annotations. Result of blastx analysis: 179944 sequences of 46.99%, 133455 (34.85%), 179978 (47.00%), respectively.

Annotation statistics	
Number of blastx results	
NCBI nr	179944 (46.99 %)
Swiss-Prot	133455 (34.85 %)
TrEMBL	179978 (47.00 %)

Assembly Results
The trimmed reads, used for the de novo transcriptome assembly, were 441,428,407. The final transcriptome, the set of unigenes, comprised a total of 1295741 transcripts with an N50 of 914 bp. The completeness for BUSCO evaluation was above 98.4%.

	Trinity	rnaSPAdes	CD-HIT-est (Unigens)
TransRate v.1.0.3			
Total transcripts	1580186	1244846	1295741
N50	684	1475	914
GC content (%)	44.00	44.00	44.00
BUSCO v.3.0.2			
Complete BUSCOs (C)	253 (99.2%)	247 (96.8 %)	251 (98.4 %)
Fragmented BUSCOs (F)	1 (0.4 %)	5 (2.0 %)	2 (0.8 %)
Missing BUSCOs (M)	1 (0.4%)	3 (1.2 %)	2 (0.8 %)